

Institute of Actuaries of India

Subject CT4 – Models

April 2016 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**Benefits:**

- Systems with long time frames can be studied in compressed time. Complex systems with stochastic elements can be studied (especially by simulation modelling in case of Cat Models).
- Different future policies or possible actions can be compared either to see which best suits the requirements of a user or to examine different scenarios without carrying them out in practice, or to avoid potential costs associated with trialling in real life.
- Models allow control over experimental conditions, so that we can reduce the variance of the results output without upsetting the mean values. Parameters can be sensitivity tested using a model.

Limitations

- Model development requires a lot of time and expertise, and hence can be /costly. May need to run model lots of times (essential if it is a stochastic model).
- Models more useful for comparing the results of input variations than for optimising outputs.
- Models can look impressive, but can lull the user into a false sense of security.
- Impressive output is not a substitute for validity and close imitation of the real world.
- Models rely heavily on the data input. If this is poor or lacking in credibility the output is likely to be flawed.
- Users need to understand the model sufficiently well to be able to know when it is appropriate to apply it.
- Interpretation of models can be difficult, and often outputs need to be seen in relative rather than absolute terms.
- Models cannot take into account all possible future events (e.g. changes in legislation).

[4 Marks]**Solution 2:**

- i) **Strict stationary:** For a process to be strictly stationary, the joint distribution of $X_{t_1}, X_{t_2}, \dots, X_{t_n}$ and $X_{t+1}, X_{t+2}, \dots, X_{t+n}$ are identical for all $t, t_1, t_2, \dots, t_n$ in J and all integers n . This means that the statistical properties of the process remain unchanged over time.

[1.5]

- ii) **Weak Stationary:** Because strict stationarity is difficult to test fully in real life, we also use the less stringent condition of weak stationarity. Weak stationarity requires that the mean of the process, $E[X_t] = m(t)$, is constant and the covariance, $E[(X_s - m(s))(X_t - m(t))]$, depends only on the time difference $t - s$.

[1.5]

- iii) **Mixed process:** It is a stochastic process that operates in continuous time, which can also change value at predetermined discrete instants. The number of contributors to a pension scheme can be modelled as a mixed process with state space $S = \{1, 2, 3, \dots\}$ and time interval $J = [0, \infty]$.

[1.5]

- iv) **Counting process** Is a process, X , in discrete or continuous time, whose state space is the natural numbers $\{0, 1, 2, \dots\}$. $X(t)$ is a non-decreasing function of t . Number of claims reported to an insurer by time t .

[1.5]

[6 Marks]

Solution 3:

i)

 $\hat{S}_{KM}(t)$ for Group A =1 for $0 \leq t < 1$ 0.87 for $1 \leq t < 2$ 0.80 for $2 \leq t < 4$ 0.65 for $4 \leq t < 5$ 0.51 for $5 \leq t < 6$ 0.36 for $6 \leq t < 7$ 0.27 for $7 \leq t < 10$ 0.14 for $t \geq 10$ $\hat{S}_{KM}(t)$ for Group B =1 for $0 \leq t < 1$ 0.73 for $1 \leq t < 2$ 0.53 for $2 \leq t < 3$ 0.33 for $4 \leq t < 4$ 0.20 for $5 \leq t < 5$ 0.07 for $6 \leq t < 6$ 0 for $t \geq 6$

The patients provided with Treatment A have a longer time to remission than Treatment B indicating that Treatment A is more effective than B. [5]

- ii) The median for Group A is 6 months and for group B is 3 months

[1]

[6 Marks]

Solution 4:

- i) Let N_t denote the number of claims up to time t . Since the Poisson process has stationary increments, we may take $t = 0$, so that the required conditional distribution is

$$\begin{aligned} P(T_0 \leq y | N_s = 1) &= P(T_0 \leq y, N_s = 1) / P(N_s = 1) \\ &= P(N_y = 1, N_s - N_y = 0) / P(N_s = 1) \end{aligned}$$

But $N_s - N_y$ is independent of N_y and has the same distribution as N_{s-y} . Thus the right hand side above equals

$$\frac{(\lambda y \exp(-\lambda y)) * (\exp(-\lambda(s-y)))}{\lambda s \exp(-\lambda s)} = y/s$$

which is the cdf of the uniform distribution on $[0, s]$.

[3]

- ii) Since holding times are independent, each having an exponential distribution, their joint density is

$$\lambda^n \exp(-\lambda(t_1 + t_2 + \dots + t_n)) \mathbf{1}_{\{t_1, t_2, \dots, t_n > 0\}}$$

[1]

- iii) We have, as in part i),

$$\begin{aligned} P(N_s = k | N_t = n) &= P(N_s = k, N_t = n) / P(N_t = n) \\ &= P(N_s = k, N_t - N_s = n - k) / P(N_t = n) \end{aligned}$$

Using again that the Poisson process has stationary and independent increments, and that the number of claims in an interval $[0, t]$ is Poisson (λt) , we derive from above that

$$\begin{aligned} P(N_s = k | N_t = n) &= (\exp(-\lambda s) * (\lambda s)^k * \exp(-\lambda(t-s)) * \lambda^{n-k} (t-s)^{n-k}) / k! * (n-k)! / (\exp(-\lambda t) * (\lambda t)^n / n!) \\ &= \exp(-\lambda t) * (\lambda^n s^k) * (t-s)^{n-k} / k! * (n-k)! * n! / (\exp(-\lambda t) * (\lambda t)^n) \\ &= n! / (n-k)! * s^k (t-s)^{n-k} / t^k * t^{n-k} \\ &= \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k} \end{aligned}$$

which is binomial with parameters n and s/t .

[3]

[7 Marks]

Solution 5:

- i) A proportional hazard model is a regression model in which the hazard rate for any specific individual is expected to be proportional to the hazard rate for a baseline individual.

The formula for a proportional hazard model defining hazard function for the i th life is

$$\lambda(t, z_i) = \lambda_0(t)g(z_i)$$

where $\lambda_0(t)$ is a function only of duration t , and $g(z_i)$ is a function only of the covariate vector.

$\lambda_0(t)$ is the hazard for an individual with a covariate vector equal to zero and is called the baseline hazard.

[2]

- ii) In a fully parametric model there is a strong assumption that the lifetime distribution and hence the hazard belongs to a given family of parametric distributions, and the regression problem is reduced to estimating the parameters from the data.

Distributions commonly used are the exponential (constant hazard), Weibull (monotonic hazard), Gompertz-Makeham (exponential hazard) and log-logistic ("humped" hazard).

Parametric models can be used with a homogeneous population.

However fully parametric models are difficult to apply without foreknowledge of the form of the hazard function. Moreover, in many applications the impact of the covariates is more important to determine than the baseline hazard. Hence semi-parametric approaches where the determination of the parameters of the covariate vector is focused on are used.

[2]

- iii) The partial likelihood gives the likelihood for the observed deaths. Since 4 deaths have been observed the partial likelihood will be the product of 4 terms.

The contribution

From the 1st death is "L1" =

$$\frac{\mu_{56}}{\mu_{56}} * \frac{e(\beta_1 + 35\beta_2 + \beta_3)}{\sum e(\beta z)}$$

Where $\sum e(\beta z) = e(\beta_1 + 35\beta_2 + \beta_3) + e(\beta_1 + 25\beta_2 + 2\beta_3) + 2e(30\beta_2 + \beta_3) + e(\beta_1 + 35\beta_2 + 2\beta_3) + e(35\beta_2 + \beta_3) + e(\beta_1 + 30\beta_2 + 3\beta_3) + e(30\beta_2 + 2\beta_3) + e(25\beta_2 + 4\beta_3) + e(30\beta_2 + 3\beta_3)$

From 2nd death is "L2" =

$$\frac{\mu_{58}}{\mu_{58}} * \frac{e(30\beta_2 + \beta_3)}{\sum e(\beta z)}$$

$$\sum e(\beta z) = 2e(30\beta_2 + \beta_3) + e(\beta_1 + 35\beta_2 + 2\beta_3) + e(35\beta_2 + \beta_3) + e(\beta_1 + 30\beta_2 + 3\beta_3) + e(30\beta_2 + 2\beta_3) + e(25\beta_2 + 4\beta_3) + e(30\beta_2 + 3\beta_3)$$

From 3rd death is "L3" =

$$\frac{\mu_{60}}{\mu_{60}} * \frac{e(35\beta_2 + \beta_3)}{\sum e(\beta z)}$$

$$\sum e(\beta z) = e(30\beta_2 + \beta_3) + e(35\beta_2 + \beta_3) + e(\beta_1 + 30\beta_2 + 3\beta_3) + e(30\beta_2 + 2\beta_3) + e(25\beta_2 + 4\beta_3) + e(30\beta_2 + 3\beta_3)$$

(Assuming that the censoring of the life at age 60 happens just after the death at age 60)

From 4th death is "L4" =

$$\frac{\mu_{63}}{\mu_{63}} * \frac{e(30\beta_2 + 2\beta_3)}{\sum e(\beta z)}$$

$$\sum e(\beta z) = e(30\beta_2 + 2\beta_3) + e(25\beta_2 + 4\beta_3) + e(30\beta_2 + 3\beta_3)$$

The partial likelihood is a product of all the terms and is

$$L = L_1 * L_2 * L_3 * L_4$$

[5]

- iv) The parameters would be estimated by maximizing the partial likelihood. Since there are more than 1 parameter to be estimated simple method like maximizing the log-likelihood would not work. The maximization can be done by iterative numerical techniques like Newton-Raphson method, which uses repeated calculations to refine the choice of regression parameters.

[1]

- v) The hazards being proportional the ratio is

$$e(25 * \beta_2 + 3 * \beta_3) / e(35 * \beta_2 + x * \beta_3), \text{ where } x \text{ is the exercise score of 2}^{\text{nd}} \text{ life}$$

Substituting the values of β_2 and β_3 , x can be derived as 4.

[2]

[12 Marks]

Solution 6:

- i) The parameter of Poisson distribution is same as mean of the distribution and estimated as "Total number of deaths" / Number of days

$$\lambda = \text{Number of deaths} / \text{Total number of days}$$

$$= 873/1826 = 0.4781$$

[1.5]

- ii) Assuming that the number of deaths follows a Poisson distribution with mean λ , the probability of 'n' deaths is given by

$$P(N = n) = \frac{e^{-\lambda} \lambda^n}{n!}$$

Using the above formula, expected deaths is derived and deviation of actual from expected tabulated

No of deaths	Expected no of days	Actual no of days	(A-E) ² /E
0	1132.73	1134	0.0014
1	540.88	540	0.0014
2	129.13	126	0.0761
3	20.55	23	0.2911
4	2.45	3	0.1217
Total			0.4917

There are 5 groups with one parameter estimated hence the degrees of freedom is 4.

Value of χ^2 is 0.4917 which is lower than the critical value of chi-squared distribution with 4 degrees of freedom at 5 percent (9.488)

Hence the Poisson distribution is an appropriate model for the deaths observed.

[2.5]

- iii) The probability of observing 6 deaths is

$$\frac{e^{-\lambda} \lambda^6}{6!} \text{ which is } \mathbf{0.001021\%}. \quad [1]$$

- iv) The parameter of Poisson distribution is same as mean of the distribution and estimated as "Total number of deaths" / Number of days

$$\lambda = \text{Number of deaths} / \text{Total number of days}$$

$$= 2782/1826 = 1.5235 \quad [1]$$

- v) Assuming that the number of deaths follows a Poisson distribution with mean λ , the probability of 'n' deaths is given by

$$P(N = n) = \frac{e^{-\lambda} \lambda^n}{n!}$$

Using the above formula, expected deaths is derived and deviation of actual from expected tabulated

No of deaths	Expected no of days	Actual no of days	(A-E) ² /E
0	79.55	67	1.9791
1	121.19	119	0.0397
2	92.32	94	0.0305
3	46.89	54	1.0795
4	17.86	21	0.5527
5-6	6.82	10	1.4795
Total			5.161

There are 6 group with one parameter estimated hence the degrees of freedom is 5.

Value of χ^2 is 5.161 which is lower than the critical value of chi-squared distribution with 5 degrees of freedom at 5 percent (11.07)

Hence the estimated parameter does give a good fit for the observed deaths.

[2.5]

- vi) From the deaths in different years, it is seen that there is a change in the distribution observed in year 2014. There is sharp increase in the number of days with 2 or more deaths. Hence a better approach would be to assume that the distribution in year 2014 and 2015 are similar and estimate the parameter accordingly

$$\lambda = \text{Number of deaths} / \text{Total number of days}$$

$$= 1225/730 = 1.6781$$

Using the above estimated parameter, the expected deaths for year 2014 is derived and the goodness of fit checked as below

No of deaths	Expected no of days	Actual no of days	(A-E) ² /E
0	65.68	67	0.4916
1	112.65	119	0.3580
2	96.60	94	0.0596
3	55.23	54	0.0109
4	23.68	21	0.3031
5	8.12	8	0.1551
6	2.32	2	0.1982
Total			1.5763

Value of χ^2 is 1.5763 which is much lower than the critical value of chi-squared distribution with 6 degrees of freedom at 5 percent (12.592). Hence the alternate approach wherein the distribution of deaths in year 2014 and 2015 is different from the earlier years gives a good fit.

[3.5]

[12 Marks]

Solution 7:

i)

	Date of birth	Date of child birth	Remarks	Exposure
A	1.1.1980	1.11.2014	Died on 1st May 2015	4 Months
B	1.2.1980	1.2.2015	Policy lapsed on 31st Jan 2015	Nil
C	1.4.1980	1.12.2014	Died on 1st July 2015	3 Months
D	1.10.1981	1.1.2015	-	Nil
E	1.5.1979	1.1.2014	-	8 Months

[2]

ii)

- a) The rate interval implied by the definition is a life year interval.

[1]

- b) The age definition is age last birthday as implied above and hence is a life year rate interval. If age were to be defined as age last birthday on the preceding '1st April, then the rate interval is a calendar year rate interval.

[1]

iii)

- a) The age definition relates to the age next birthday on 1st January and hence is a calendar year rate interval.

The age at the start of the interval is (x-1, x)

[2]

- b) The deaths would be captured as deaths related to age 'x' if year of death less year of birth is equal to x.

The rate interval is a calendar year interval and is defined as 'x' next birthday on 1st January 2013 or age 'x-1' last birthday on 1st January 2013. The corresponding exposed to risk, assuming birthdays are uniformly distributed across the calendar year, is average of lives aged 'x' next birthday at the beginning of the calendar year (1st January 2013) and lives aged 'x+1' next birthday at the end of the calendar year (1st January 2014). Using census approximation, this can be approximated by the population aged 'x' nearest birthday on 1st July 2013 which is provided in the data, i.e 2013 less year of birth will give age 'x' nearest birthday.

[3]

[9 Marks]**Solution 8:**

$$\begin{aligned} \text{i) } d/dt P_{FF}(t) &= -2t * P_{FF}(t) \\ d/dt(\ln P_{FF}(t)) &= -2t \end{aligned}$$

$$\ln P_{FF}(s) = -s^2 + \text{constant}$$

We Know,

$P_{FF}(0) = 1$, hence constant = 0

$$P_{FF}(s) = \exp(-s^2)$$

[3]

ii) $P(\text{in first visit to U at time } T \text{ in state F at } t = 0)$

$$\int_0^T P(\text{remains in F to times } s) * P(\text{transition to U in time } s, s + ds) * P(\text{remains in S to time } T) ds = \int_0^T P_{FF}(s) * 2s * P_{UU}(s, T) ds$$

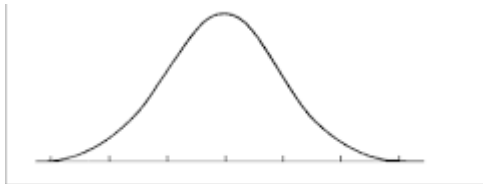
Using the result from part (i) and the similar result for P_{BB} with boundary condition $P_{UU}(s, s) = 1$, this gives us:

$$\begin{aligned} & \int_{s=0}^T \exp(-s^2) * 2s * \exp(-T^2 + s^2) ds \\ &= \int_{s=0}^T \exp(-T^2) * 2s ds \\ &= \exp(-T^2) * T^2 \end{aligned}$$

[4]

iii)

a) The sketch should be shaped like:



[2]

b) Commentary:

Initially probability increases from 0 at $T = 0$, and accelerates as the transition rate from F to U increases. However, as transitions increase, it becomes more likely that the process has already visited state U and jumped back to F. Therefore the probability of being in the first visit to U tends (exponentially) to zero.

[2]

c) Differentiate to find turning point:

$$\begin{aligned} d/dt(\exp(-t^2) * t^2) &= 2t * \exp(-t^2) - 2t^3 * \exp(-t^2) \\ \text{set derivative equal to zero} \\ \exp(-t^2) * 2t(1 - t^2) &= 0 \end{aligned}$$

implies $t = 1$ for a positive solution and, from above analysis, this is clearly a maximum.

[2]

[13 Marks]

Solution 9:

- i) While the chi-squared test is a good test for the overall goodness of fit, it fails to detect several defects that could be of considerable financial importance.
- a. There could be a few large deviations offset by a lot of very small deviations. The chi-squared test could be satisfied although the data do not satisfy the distributional assumptions that underlie it. This is, because the chi-squared -statistic summarises a lot of information in a single figure.
 - b. The graduation might be biased above or below the data by a small amount. The statistic can often fail to detect consistent bias if it is small, but a small bias would have financial impact, if mortality is underestimated (in life insurance) or overestimated (In pensions and annuity) .
 - c. Even if the graduation is not biased as a whole, there could be significant groups of consecutive ages (called runs or clumps) over which it is biased up or down.
 - d. As test is based on squared deviations, it tells us nothing about the direction of any bias or the nature of any lack of adherence to data of a graduation, even if the bias is large or the lack of adherence manifest.
- [2]
- ii) The following tests can be used to detect the flaws in chi-squared test
- a. Standardised deviations test: The test detects a number of large deviations. If the graduated rates are not a good fit, the distribution will not be 'centred' correctly.
 - b. Signs test: The signs test is a simple test for overall bias, *ie* whether the graduated rates are too high or too low. This test will identify the failure of chi-squared test to detect where there is an imbalance between positive and negative deviations.
If the graduated rates do not tend to be higher or lower than the crude rates on average, we would expect roughly half the graduated values to be above the crude rates and half below. So, if there are m age groups, the number above (or below) should have a $B(m, \frac{1}{2})$ distribution. An excessively high number of positive or negative deviations will indicate that the rates are biased.
 - c. Grouping of signs test: It detects grouping of deviations of the same sign. The test looks at the number of groups (or runs) of deviations of the same sign and compares this with the number that would be expected if the positive and negative signs were arranged in random order. If the graduated rates are overgraduated, the standardised deviations will not swap from positive to negative very often and there will be fewer runs than expected
 - d. Cumulative deviations: The cumulative deviations test detects overall bias or long runs of deviations of the same sign. It addresses the problem of the inability of the chi square test to detect a large positive or negative cumulative deviation over part (or the whole) of the age range.
 - e. Serial correlations test : This test detects grouping of signs of deviations. If the graduates rates are over-graduated the curve tends to stay on the same side of the crude rates for relatively longer periods and the consecutive

deviations tend to be positively correlated. The correlation coefficient of lagged sequences is computed and a positive value indicates that nearby values of Z_x tend to have similar values and negative value indicates that nearby values of Z_x tend to have opposite values.

[2]

iii)

- (a) Standardised deviations test: The observations of Z_x are divided into 6 groups $(-\infty, -2)$, $(-2, -1)$, $(-1, 0)$, $(0, 1)$, $(1, 2)$, $(2, \infty)$. The expected number of Z_x that should be in each group is compared with the actual numbers

Range	$(-\infty, -2)$	$(-2, -1)$	$(-1, 0)$	$(0, 1)$	$(1, 2)$	$(2, \infty)$
Expected	0.4	2.8	6.8	6.8	2.8	0.4
Actual	2	1	9	4	2	2
$(A - E)^2/E$	6.4	1.2	0.7	1.2	0.2	6.4

The χ^2 statistic will have a χ^2 distribution with 5 degrees of freedom (as 6 intervals are used and we have constrained the total to 20).

The observed value of 12.3109 is higher than the critical value of 11.070 and hence the graduation is not a good fit.

- (b) Signs test

The number of positive signs is 8 and negative is 12. The number of signs is Binomial $(20, 1/2)$. The probability of obtaining 8 positive signs in 20 observations is $2 * 0.2517$, which is higher than 5%. Hence there is no evidence to reject the graduated values.

- (c) Grouping of signs test : The number of positive deviations is 8 and negative is 12. From the tables, the critical value when n_1 is 8 and n_2 is 12 is 2. Since we have observed 2 groups of positive deviations, we reject the graduated rates and conclude there is evidence of grouping of deviations of the same sign,

[5]

- iv) The number of policies based on the distribution is

Assume that n_1 lives have 1 policy n_2 lives have 2 policies and so on.

For lives, the death rate = deaths/lives = 0.005

For policies we have death rate = claims/policies

$$= (0.005 n_1 + 0.005 * 2(n_2) + 0.005 * 3(n_3)...) / (n_1 + 2(n_2) + 3(n_3)...) = 0.005$$

$$= 0.005 * (n_1 + 2(n_2) + 3(n_3)...) / (n_1 + 2(n_2) + 3(n_3)...) = 0.005$$

Hence proved

[2]

v)

Variance ratio is given by

$r = \frac{\sum_i i^2 \pi_i}{\sum_i i \pi_i}$, where π_i represents the proportion of policies in each group with i policies.

$$r = \frac{3.506}{1.418} = 2.4725$$

[2]

[13 Marks]**Solution 10:**

i) The generator matrix of the process $\{\sigma_{ij} : i, j = A, F, M, S, O\}$, would be

$$\begin{array}{ccccc} -1 & 0.4 & 0.1 & 0.5 & 0 \\ 0 & -1/3 & 1/12 & 1/12 & 1/6 \\ 0 & 0 & -1/60 & 0 & 1/60 \\ 0 & 0 & 0 & -1/2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{array}$$

[3]

ii) The probability of ever visiting state M is $= 1/10 + 4/10 * 1/4 = 1/5$

[1]

iii)

a) $d/dt P_{AA}(t) = -P_{AA}(t)$, which has solution $P_{AA}(t) = \exp(-t)$

[1.5]

b) Similarly,

$$d/dt P_{AF}(t) = (-1/3)P_{AF}(t) + 0.4P_{AA}(t), \text{ so that}$$

$$\begin{aligned} d/dt(\exp(t/3) P_{AF}(t)) &= 0.4 \exp(t/3) P_{AA}(t) = 0.4 \exp(-2t/3) \\ \text{giving } P_{AF}(t) &= \exp(-t/3) * 0.6 * (1 - \exp(-2t/3)) \end{aligned}$$

[1.5]

iv)

a) The equation arises as follows: when the process arrives in state i the subsequent holding time has mean $1/\lambda_i$ after which the process jumps to a different state, choosing state j with probability $p_{ij} = \sigma_{ij} / \lambda_i$ (independent of the length of the holding time). The total time to reach

state O is therefore the time until the first jump plus the time from arriving in the new state until hitting state O.

[2]

b) We have $m_m = 60$, $m_s = 2$, $m_f = 3 + \frac{1}{4} \cdot 60 + \frac{1}{4} \cdot 2 = 18.5$
 $M_A = 1 + 0.1 \cdot 60 + 0.5 \cdot 2 + 0.4 \cdot 18.5 = 15.4$ hours.

[2]

v) The time-homogeneous Markov model has exponential holding times, so the distribution is completely determined by the expectation.

[1]

vi) A simple check on whether the Markov model fits the data is therefore to verify that the distributions of holding times are at least roughly exponential, and a simple way of doing that is to compare sample standard deviations with sample means. More detailed comparisons might be possible, depending on the size of the data set.

[2]

vii)

a) Calculations required in the first case would include working out the expected duration of stay if the change were implemented, which involves solving the equations in (iv) again. For the second situation, just replace m_s in the original calculation. New parameter values will need to be guessed. Whichever model comes out better should be compared with the initial situation, to determine whether the improvement was worth the additional resources.

[2]

b) Model suitability: on the one hand the required decision is couched in terms of expectations, which lend themselves well to Markov process treatment. On the other, the fundamental problem in the system is queue length, which can never be successfully modelled by a process which tracks only a single individual at a time. (A network of queuing processes would be a much better model.)

[2]

[18 Marks]
